

Assignment - 04

Q Convert the following:

- (i) $(1AD.E0)_{16} = (?)_{10}$
 (ii) $(37.625)_{10} = (?)_2$
 (iii) $(110100111101.110)_2 = (?)_8$
 (iv) $(345.AB)_{16} = (?)_{10}$

(1) $(1AD.E0)_{16}$ to Decimal $(?)_{10}$.

$$\rightarrow 1 \times 16^2 + A \times 16^1 + D \times 16^0 + E \times 16^{-1} + 0 \times 16^{-2}$$

$$\rightarrow 1 \times 256 + 10 \times 16 + 13 \times 1 + 14 \times 0.0625 + 0 \times 0.0039$$

$$\rightarrow 256 + 160 + 13 + 0.875$$

$$\rightarrow 429.875_{10}$$

(2) $(37.625)_{10}$ to Binary $(?)_2$ $\rightarrow$ Convert 37 to binary:

$$2 \overline{) 37} - 1$$

$$2 \overline{) 18} - 0$$

$$2 \overline{) 9} - 1$$

$$2 \overline{) 4} - 0$$

$$2 \overline{) 2} - 0$$

$$2 \overline{) 1} - 1$$

$$\Rightarrow (100101)_2$$

$$0.625 \times 2 = 1.25 \rightarrow 1$$

$$0.25 \times 2 = 0.5 \rightarrow 0$$

$$0.5 \times 2 = 1 \rightarrow 1$$

$$\Rightarrow (0.101)_2$$

Final answer $(100101.101)_2$.(3) $(110100111101.110)_2$ to Octal $(?)_8$.

• Grouping binary digits

$$\therefore (6475.6)_8$$

$$\rightarrow 110 \ 100 \ 111 \ 101 \ . \ 110$$

$$\Rightarrow 6 \ 4 \ 7 \ 5 \ . \ 6$$

4) $(3+5.AB)_{10}$ to binary.
 • Convert each hex digit to 4-bit binary.

→ 3 → 0011

→ 4 → 0100

→ 5 → 0101

→ A → 1010

→ B → 1011

Final answer: $(001101000101 \cdot 10101011)_2$

2) Mention any three theorems of Boolean's algebra and prove each of them.

⇒ Boolean theorem

1) $A + (B + C) = (A + B) + C$

2) $A(BC) = (AB)C$

3) $A(B + C) = AB + AC$

(i) $A + (B + C) = (A + B) + C$

A	B	C	$B + C$	$A + (B + C)$	$A + B$	$(A + B) + C$
0	0	0	0	0	0	0
0	0	1	1	1	0	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Hence $A + (B + C) = (A + B) + C$

$$2) A(BC) = (AB)C$$

A	B	C	BC	A(BC)	AB	(AB)C
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	1	1	1	1

Hence $A(BC) = (AB)C$

$$3) A(B+C) = AB + AC$$

A	B	C	B+C	A(B+C)	AB	AC	AB+AC
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

Hence $A(B+C) = AB + AC$

3) using basic boolean theorems, Prove

$$(i) (x+y)(x+yz) = x + yz$$

$$(ii) xy + xz + yz = xz + yz$$

$$\begin{aligned}(i) \text{ LHS} &= (x+y)(x+xz) = xx + xz + xy + yx \\&= x + xz + xz + yz \quad (xx = x) \\&= x(1+z) + yz \\&= x \cdot 1 + xz + yz \quad [(1+z) = 1] \\&= x + xz + yz \quad (1+z = 1) \\&= x + yz, \\&\text{Hence, proved.}\end{aligned}$$

$$\begin{aligned}(ii) \text{ LHS} &= xy + xz + yz \\&= xy + xz + yz \\&= xy + xz + (x+x')yz \\&= xy + xz + xy\bar{z} + x'y\bar{z} \\&= (xy + xy\bar{z}) + (xz + x'y\bar{z}) \\&= y\bar{z}(1+x) + xz(1+y) \\&= y\bar{z}(1) + xz(1) \\&= y\bar{z} + xz\end{aligned}$$

(4) Express the boolean function $F = A + B'c$ in a sum of minterms

$\Rightarrow$ The function has three variables A, B, c . The first term A is missing two variables, $F = A + B'c \rightarrow (1)$

$$\therefore A = A(B+B') = AB + AB'$$

This is still missing one variable

$$A = AB(c+c') + AB'(c+c')$$

$$A = ABC + ABC' + AB'c + AB'c' \rightarrow (2)$$

The second term $B'C$ is missing one variable,
 $B'C = B'C(A + A') = AB'C + A'B'C \rightarrow \textcircled{2}$

Sub in $\textcircled{2}$ & $\textcircled{3}$ in $\textcircled{1}$.

$$\textcircled{1} \Rightarrow F = ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C \quad A+A=A$$

$$F = ABC + ABC' + AB'C + AB'C' + A'B'C$$

$$F = m_1 + m_2 + m_5 + m_6 + m_7$$

$$F(A, B, C) = \sum (1, 2, 5, 6, 7)$$

Truth table: $f = A + B'C$

A	B	C	f
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$\textcircled{5}$ Express the Boolean function $F = xy + \bar{x}z$ in a product of maxterms form

$$F = xy + \bar{x}z \rightarrow \textcircled{1}$$

The given function is in sum of product, so first convert the function into OR terms using distributive law:

$$F = xy + \bar{x}z = (xy + x')(xy + z)$$

$$= (x + \bar{x})(y + \bar{x})(x + z)(y + z)$$

$$F = (\bar{x} + y)(x + z)(y + x) \rightarrow \textcircled{2}$$

The function has three variables x, y and z .
 Each or terms is missing one variable.

$$\bar{x}y = \bar{x} + y + z\bar{z} = (\bar{x} + y + z)(\bar{x} + y + \bar{z})$$

$$x + z = x + z + yy' = (x + z + y)(x + z + y')$$

$$y + x = y + z + x\bar{x} = (x + y + z)(\bar{x} + y + x)$$

combining all the terms and remaining those that appear more than once, we obtain

$$F = (x + y + z)(x + \bar{y} + z)(\bar{x} + y + z)(\bar{x} + y + \bar{z})$$

$$F = M_0 M_2 M_4 M_5$$

$$\therefore F(x, y, z) = \pi(0, 2, 4, 5)$$

6) Explain SOP and POS with example

⇒ SOP

- Sum of products
 - SOP is boolean function containing AND terms called product terms. of one or more literals each
 - The sum denote the bring of these terms.
- Function expressed in sum of product is

$$F_1 = y' + xy + x'yz'$$

$$\text{ex: } AB + BC + AC$$

⇒ POS

- product of sums
 - POS is a boolean function contain OR terms called sum terms
 - Each term may have any number of literals
 - The product denotes the ANDing of these terms
- function expressed in sum of product is

$$F_2 = x(y' + z)(x' + y + z' + w)$$

$$\text{ex: } y = (A + B + C) \cdot (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C})$$

7) what are logic gates & write the graphic symbol, algebraic function & truth table of all 8 logic gates.

→ A logic gate is a digital device that performs a boolean function a logical operation performs on one or more binary inputs that produce a single binary output.

① Buffer gate



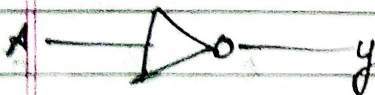
logic equation

$$[A = Y]$$

Truth table

A	Y
0	0
1	1

② Not gate



logic equation

$$[Y = \bar{A}]$$

Truth table

A	$\bar{Y}$
0	1
1	0

③ AND gate



logic equation

$$Y = A \cdot B$$

Truth table.

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

④ OR gate



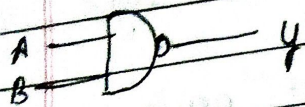
expression

$$Y = A + B$$

Truth table.

A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

5) NAND gate



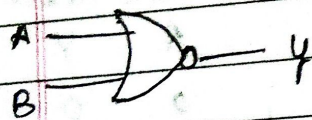
Expression

$$Y = \overline{A \cdot B}$$

T.T

A	B	A.B	$\overline{A.B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

6) NOR gate



Expression

$$Y = \overline{A + B}$$

T.T

A	B	A+B	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

7) X-OR gate



Expression

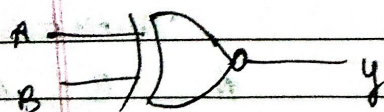
$$Y = \overline{A}B + A\overline{B}$$

$$Y = A \oplus B$$

T.T

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

8) X-NOR gate



Expression

$$Y = \overline{A}B + A\overline{B}$$

$$Y = A \odot B$$

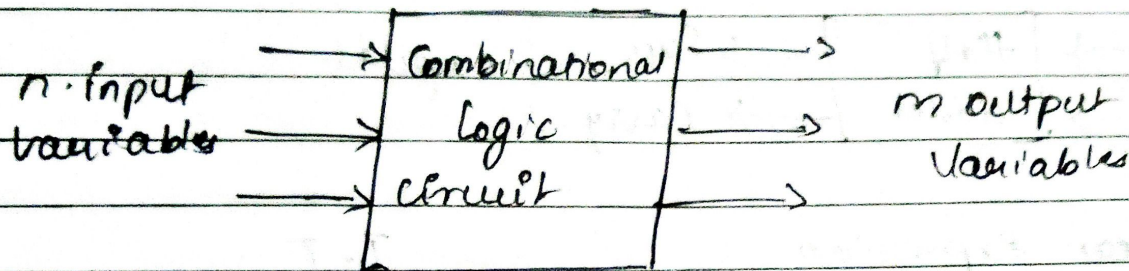
T.T

A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

write the step-by-step procedure to design a Combination Circuit.

Logic Circuits for digital systems may be Combinational or Sequential.

A combinational circuit consists of logic gates whose outputs are determined directly from present combination of inputs at any time.



Block diagram of combinational circuit.

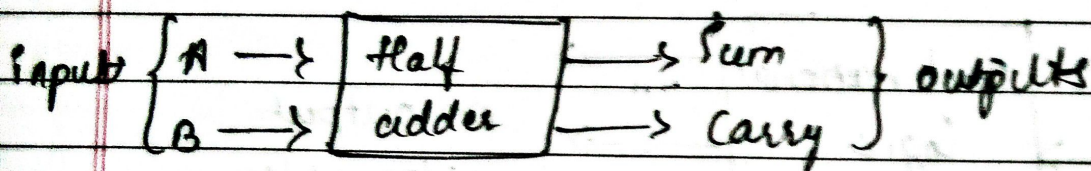
Procedure:-

1. The problem is stated.
2. The noⁿ output available input variables & required output variables are assigned letter symbols.
3. The truth table that defines the required relationship b/w inputs & outputs is derived.
4. The simplified boolean function for each output is obtained.
5. The logic diagram is drawn.

q) Design a half adder with necessary logic diagram & expressions.

- A combinational circuit that performs addition of 2 bits. These circuits need two binary inputs & 2 binary outputs.

Symbol



Boolean Expression

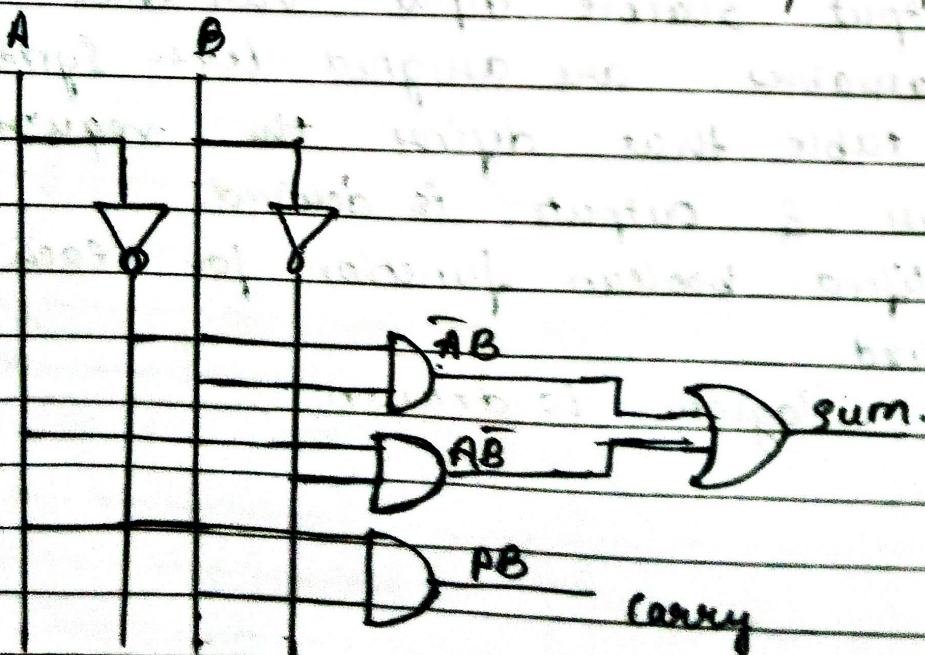
$$\text{Sum} = \bar{A}B + A\bar{B} = A \oplus B$$

$$\text{Carry} = AB$$

T.T

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

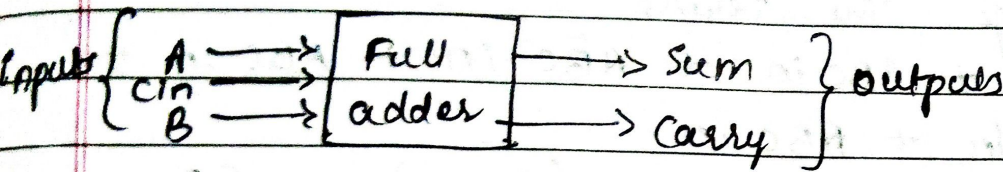
Logic diagram.



10) with the help of truth table, explain full adder using logic gates

Full adder is a combinational circuit that performs the arithmetic sum of input bits.

Block diagram:-



- this consists of 3 inputs & 2 outputs
- 2 input variables denoted by A & B represents the 2 significant bits to be added.
- 3rd input C_{in} represents the carry from the previous lower significant position

T.T

A	B	C _{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

using basic gates

$$\text{Sum} = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$\text{Carry} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$

Add $ABC_{in} + ABC_{in}$

$$\text{Carry} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in} + ABC_{in} + ABC_{in}$$

Rearrange the terms

$$= \bar{A}BC_{in} + ABC_{in} + A\bar{B}C_{in} + ABC_{in} +$$

$$A\bar{B}C_{in} + ABC_{in}$$

$$= BC_{in}(\bar{A} + A) + AC_{in}(\bar{B} + B) + AB(\bar{C}_{in} + C_{in})$$

$$= BC_{in} + AC_{in} + AB$$

$$\boxed{\text{Carry} = AB + BC_{in} + AC_{in}}$$